

UNIT-I - LINEAR PROGRAMMINGPart - B

- 1) i) A company has 3 operational departments namely weaving, processing & packing, with capacity to produce three different types of clothes namely suiting, shirting & woollens yielding a profit of Rs. 2, Rs. 4 & Rs. 3 per meter. One meter of suiting requires 3 minutes of weaving, 2 minutes of processing & 1 minute of packing. One meter of woollen requires 3 minutes in each department. One meter of shirt required 4 minutes, 1 minute & 3 minutes in weaving, process & packing respectively. In a week, total run time of each department is 60, 40 & 80 hours. Formulate as an LPP. Can this be solved by Graphical or Simplex method? If yes, Justify it.

Solution :

Step 1 : Objective of the problem is profit (That is maximization type) Then we need to maximize

Step 2 : let x_1 be the number of units of suiting
let x_2 be the number of units of shirting
let x_3 be the number of units of woollens yielding.

Decision variables are x_1, x_2, x_3

Resources are weaving, processing and packing.

Step 3 :

Department Types of clothes	weaving	Processing	Packing
Suiting (x_1)	3 mins	2 mins	1 min
Shirting (x_2)	4 mins	1 min	3 mins
Woolen yielding (x_3)	3 mins	3 mins	2 mins
Constraints ($\leq$)	60	40	80

Step 4 :

Formulation of LPP :

$$\text{Maximization } Z = 2x_1 + 4x_2 + 3x_3$$

(Profit)

$$\text{subject to (constraints) } 3x_1 + 4x_2 + 3x_3 \leq 60$$

$$2x_1 + 1x_2 + 3x_3 \leq 40$$

$$1x_1 + 3x_2 + 3x_3 \leq 80$$

$$\text{where } x_1, x_2, x_3 \geq 0$$

$\therefore$ Non-negativity restrictions.

- 1) ii) A person wants to decide the constituents of a diet which will fulfil his daily requirements of proteins, fats and carbohydrates at the minimum for the cost. The choice is to be made from four different types of foods. A diet for the person must contain minimum 800 units of proteins, 200 units of fats and 300 units of carbohydrates. Four types of food are available at a cost of Rs. 45, Rs. 40, Rs. 85 & Rs. 65 per unit respectively. Type-1 food contains 3 units of proteins, 2 unit of fats & 6 carbohydrates. Type-2 food contains 4 units proteins, 2 unit fats & 4 carbohydrates. Type-3 food contains 8 unit proteins, 7 units fats & 7 carbohydrates and Type-4 food contains 6 units proteins, 5 units fats & 4 carbohydrates. Formulate LPP for the given situation, Can this be solved by Graphical or Simplex method? If yes, Justify it.

Solution:

Step 1: Objective of the problem is a person wants to decide the constituents of a diet at the minimum costs.

Step 2: Let x_1 be the no. of units of Type 1

Let x_2 be the no. of units of Type 2

Let x_3 be the no. of units of Type 3

Let x_4 be the no. of units of Type 4

Decision variables are x_1, x_2, x_3, x_4

Resources are proteins, fats, carbohydrates.

step 3 :

Type \ Diet	(x_1)	(x_2)	(x_3)	(x_4)	Constraints ($\geq$)
Proteins	3	4	8	6	800
Fats	2	2	7	5	200
Carbohydrates	6	4	7	4	300

step 4 :

Formulation of LPP :

$$\text{Minimize } Z = 45x_1 + 40x_2 + 85x_3 + 65x_4$$

(cost)

$$\begin{aligned} \text{Subject to (constraints)} \quad & 3x_1 + 4x_2 + 8x_3 + 6x_4 \geq 800 \\ & 2x_1 + 2x_2 + 7x_3 + 5x_4 \geq 200 \\ & 6x_1 + 4x_2 + 7x_3 + 4x_4 \geq 300 \end{aligned}$$

$$\text{where } x_1, x_2, x_3, x_4 \geq 0$$

$\therefore$ Non-negativity restrictions.

- 2) i) A diet for a sick person must contain at least 4,000 units of vitamins, 50 units of minerals and 1,400 calories. Two foods A & B are available at a cost of Rs. 4 & Rs. 3 per unit respectively. If one of A contains 200 units of vitamins, 1 unit of mineral & 40 calories and one unit of food B contains 100 units of vitamins, 2 units of minerals & 40 calories, formulate the problem as an LP model and solve it by graphical method to find a combination of foods for the least cost.

Solution: I. Formulation of LPP

Step 1: Objective of the problem is minimize (Minimization Type)

Step 2: let x_1 be No. of units of food A
let x_2 be No. of units of food B

Decision variables are x_1, x_2

Resources are vitamins, Minerals, Calories

Step 3:

Diet	Food A (x_1)	Food B (x_2)	Constraints ($\geq$)
vitamins	200	100	4000
Minerals	1	2	50
Calories	40	40	1400

Step 4 : Formulation

Objective function is Minimize (Minimization)

$$\text{Minimize } Z = 4x_1 + 3x_2$$

$$\text{Subject to } 200x_1 + 100x_2 \geq 4000$$

$$\text{(constraints)} \quad 1x_1 + 2x_2 \geq 50$$

$$40x_1 + 40x_2 \geq 1400$$

$$\text{where } x_1, x_2 \geq 0$$

$\therefore$ Non-Negativity restrictions

II. To solve the LPP by Graphical Method

Step 1: Convert inequality into equality

$$\text{Now, Minimize } Z = 4x_1 + 3x_2$$

(cost)

$$\text{Subject to } 200x_1 + 100x_2 = 4000$$

$$\text{(constraints)} \quad 1x_1 + 2x_2 = 50$$

$$40x_1 + 40x_2 = 1400$$

$$\text{where } x_1, x_2 \geq 0$$

$$\therefore 2x_1 + 1x_2 = 40 \quad \text{--- ①}$$

$$1x_1 + 2x_2 = 50 \quad \text{--- ②}$$

$$1x_1 + 1x_2 = 35 \quad \text{--- ③}$$

Step 2 : To plot the points
on equation ①

$$\text{Put } x_1 = 0,$$

$$0 + x_2 = 40$$

$$x_2 = 40 \quad \therefore \text{The point A is } (0, 40)$$

$$\text{Put } x_2 = 0,$$

$$2x_1 + 0 = 40$$

$$2x_1 = 40$$

$$x_1 = 20 \quad \therefore \text{The point B is } (20, 0)$$

on equation ②

$$x_1 = 0,$$

$$0 + 2x_2 = 50$$

$$2x_2 = 50$$

$$x_2 = 25$$

$$\therefore \text{The point C is } (0, 25)$$

on equation ③

$$\text{Put } x_1 = 0$$

$$0 + 1x_2 = 35$$

$$x_2 = 35$$

$$\therefore \text{The point E is } (0, 35)$$

$$x_2 = 0$$

$$1x_1 + 0 = 50$$

$$x_1 = 50$$

$$\therefore \text{The point D is } (50, 0)$$

$$\text{Put } x_2 = 0$$

$$1x_1 + 0 = 35$$

$$x_1 = 35$$

$$\therefore \text{The point F is } (35, 0)$$

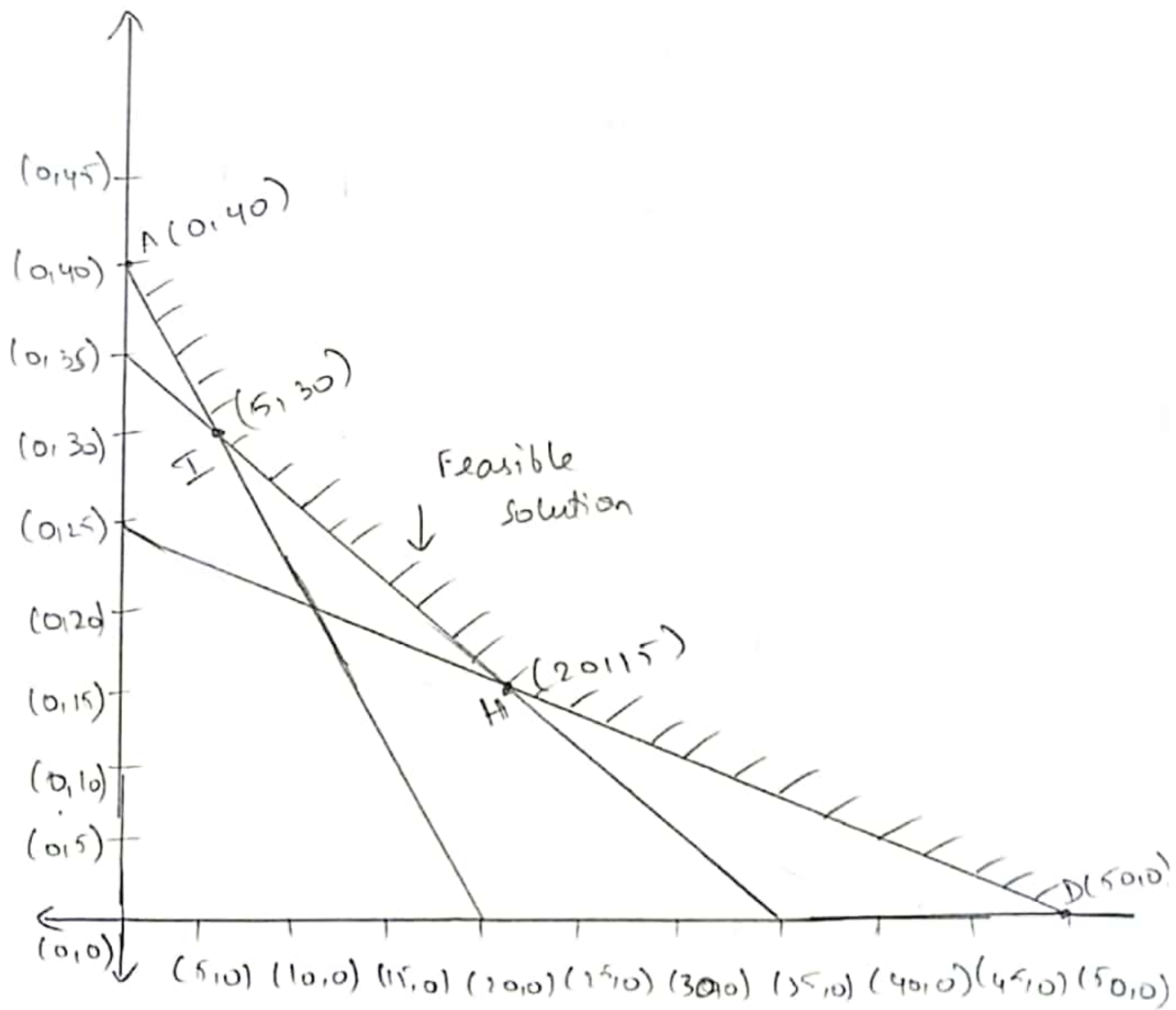
on solving ①, ② & ③

Eq ① & ② $\Rightarrow$ The point A is (10, 20)

Eq ② & ③ $\Rightarrow$ The point H is (20, 15)

Eq ③ & ④ $\Rightarrow$ The point I is (5, 30)

Step 3: Graph



Step 4 : To find the optimal solution

vertices	Objective function $\text{Min } Z = 4x_1 + 3x_2$
$A(0, 40)$	$4(0) + 3(40) = 120$
$I(5, 30)$	$4(5) + 3(30) = 110$
$H(20, 15)$	$4(20) + 3(15) = 125$
$D(50, 0)$	$4(50) + 3(0) = 200$

Step 5 : Result

$$\text{Hence } x_1 = 5, x_2 = 30$$

Minimum cost is Rs. 110

- 2ii) A company manufactures two types of products P_1 and P_2 . Each product uses lathe and milling machines. The processing time per unit of P_1 on the lathe is 5 hours and on the milling machine is 4 hours. The processing time per unit of P_2 on the lathe is 10 hours and on the milling machine 4 hours. The Maximum number of hours available per week on the lathe and the milling machine are 60 hours and 40 hours, respectively. Also the profit per unit of selling P_1 and P_2 are Rs. 6.00 and Rs. 8.00 respectively. Formulate as an LPP model and solve it graphically.

Solution:

I. Formulation of LPP

Step 1: Objective of the problem is Maximize (Maximization Type)

Step 2: Let x_1 be the no. of units of P_1

Let x_2 be the no. of unit of P_2

Decision variables are x_1, x_2

Resources are lathe and milling machine

Step 3:

Machines	Products		Constraints ($\leq$)
	$P_1 (x_1)$	$P_2 (x_2)$	
Lathe	5	10	60
Milling Machine	4	4	40

Step 4: Formulation

Objective function is maximize

$$\text{Maximize } z = 6x_1 + 8x_2$$

(profit)

$$\begin{aligned} \text{Subject to} \quad & 5x_1 + 10x_2 \leq 60 \\ \text{(Constraints)} \quad & 4x_1 + 4x_2 \leq 40 \end{aligned}$$

$$\text{where } x_1, x_2 \geq 0,$$

Non-negativity restriction

II. To solve the LPP by Graphical Method

Step 1: Convert inequality into equality

Now, maximize $Z = 6x_1 + 8x_2$

Subject to $5x_1 + 10x_2 = 60$

$4x_1 + 4x_2 = 40$

where $x_1, x_2 \geq 0$

$$\therefore x_1 + 2x_2 = 12 \text{ --- (1)}$$

$$x_1 + x_2 = 10 \text{ --- (2)}$$

Step 2: To plot the points

On equation (1)

Put $x_1 = 0$

$$2x_2 = 12$$

$$x_2 = 6$$

$\therefore$ The point A is (0, 6)

On equation (2)

Put $x_1 = 0$

$$0 + x_2 = 10$$

$$x_2 = 10$$

$\therefore$ The point C is (0, 10)

Put $x_2 = 0$

$$x_1 + 2(0) = 12$$

$$x_1 = 12$$

$\therefore$ The point B is (12, 0)

Put $x_2 = 0$

$$x_1 + 0 = 10$$

$$x_1 = 10$$

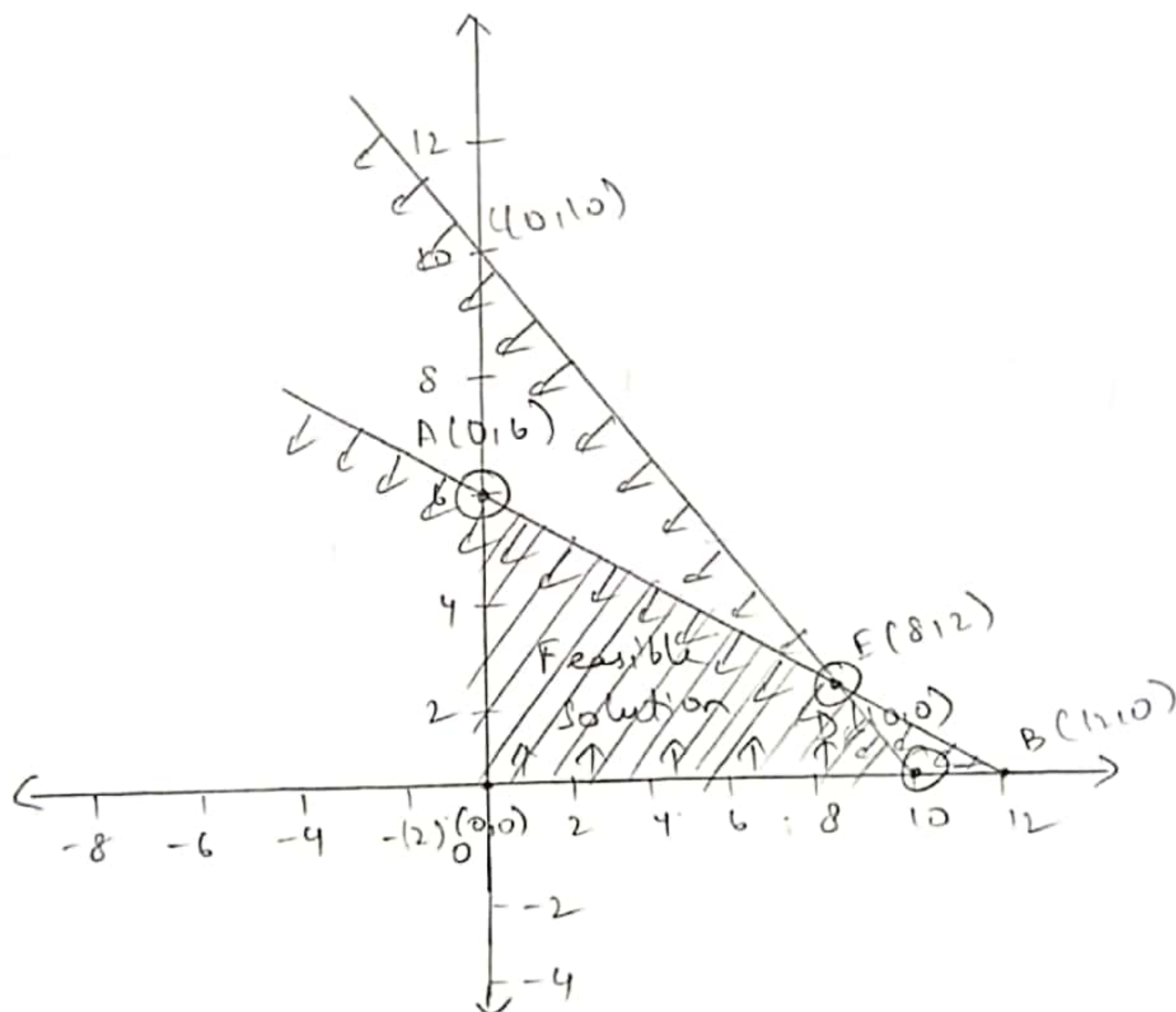
$\therefore$ The point D is (10, 0)

On solving equation ① & ②

$$x_1 = 8, \quad x_2 = 2$$

The point of Intersection is $E(8, 2)$

Step 3: Graph



Step 4: To find the optimal solution

Vertices	Objective function $\text{max } z = 6x_1 + 8x_2$
$O(0,0)$	$z = 6(0) + 8(0) = 0$
$A(0,6)$	$z = 6(0) + 8(6) = 48$
$D(10,0)$	$z = 6(10) + 8(0) = 60$
$E(8,2)$	$z = 6(8) + 8(2) = 64$

Step 5: Result

Hence, $x_1 = 8, x_2 = 2$

Maximum Profit is

Rs. 64

- 2iii) A nutritional scheme for babies is proposed by a committee of doctors. Babies can be given two types of food, I and II, which are available in standard sized packets weighing 50 grams. The cost per packet of these foods is Rs 2 and Rs 3 respectively. The vitamin availability in each type of food per packet and the minimum vitamin requirement for each type of vitamin are summarized in table. Develop a linear programming model and solve it graphically.

Vitamin	Vitamin availability per packet		Minimum daily requirement
	Food type I	Food Type II	
1	1	1	6
2	7	1	14
Cost (packet (rs))	2	3	

Solution :

I. Formulation of LPP

Step 1: Objective of the problem is to minimize the vitamin requirement [Minimization type]

Step 2: let x_1 be the food type 1

let x_2 be the food type 2

Decision variables are x_1, x_2

Resources are vitamin 1, vitamin 2

Step 3:

vitamin	Vitamin availability per packet		Minimum daily requirements	Cost / Packet (Rs)
	Food type (x_1)	Food type (x_2)		
1	1	1	6	2
2	7	1	14	3

Step 4:

Objective function is Minimize [Minimization]

$$\text{Minimize } Z = 2x_1 + 3x_2$$

(vitamin)

$$\text{Subject to } 1x_1 + 1x_2 \geq 6$$

$$\text{(constraints) } 7x_1 + 1x_2 \geq 14$$

where $x_1, x_2 \geq 0$, non-negativity constraints

II. To solve the LPP using graphical method.

Step 1: Convert inequality into equality

$$\text{Now, Minimize } Z = 2x_1 + 3x_2$$

(vitamin)

$$\text{Subject to } 1x_1 + 1x_2 \geq 6$$

$$7x_1 + 1x_2 \geq 14$$

$$\text{where } x_1, x_2 \geq 0$$

$$\therefore 1x_1 + 1x_2 = 6 \quad \text{--- (1)}$$

$$7x_1 + 1x_2 = 14 \quad \text{--- (2)}$$

Step 2 : To plot the points

On equation ①,

$$\text{Put } x_1 = 0$$

$$\Rightarrow 0 + 1x_2 = 6$$

$$x_2 = 6$$

The point A is (0, 6)

On equation ②

$$\text{Put } x_1 = 0$$

$$\Rightarrow 0 + 1x_2 = 14$$

$$x_2 = 14$$

The point C is (0, 14)

On solving eq ① & ②

The point E is $(4/3, 14/3)$

(1.3, 4.6)

$$\text{Put } x_2 = 0$$

$$1x_1 + 0 = 6$$

$$x_1 = 6$$

The point B is (6, 0)

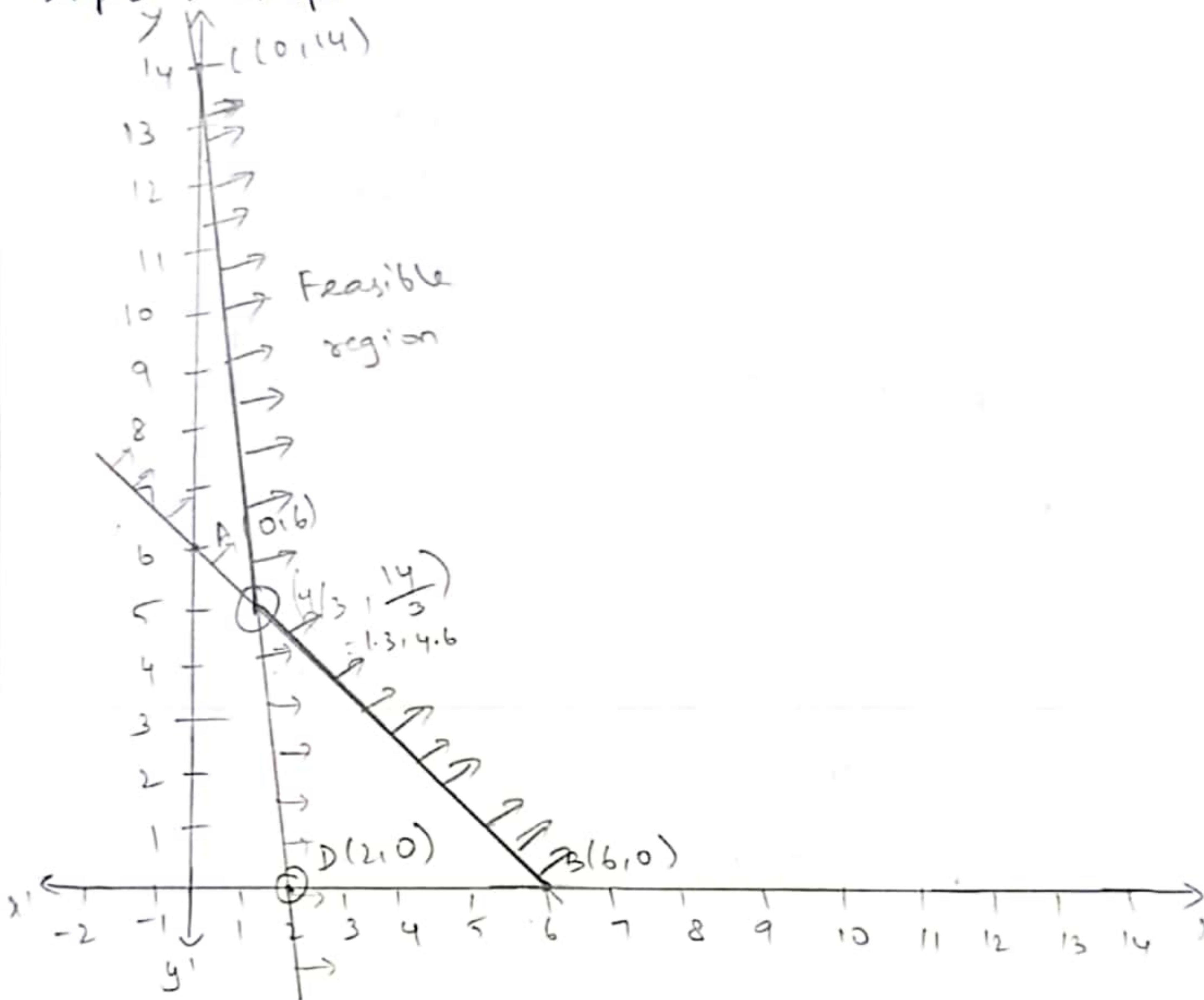
$$\text{Put } x_2 = 0$$

$$7x_1 + 0 = 14$$

$$x_1 = 2$$

The point D is (2, 0)

Step 3 : Graph



Step 4 : To find optimal solution

Vertices	Objective function $\text{Min } Z = 2x_1 + 3x_2$
C(0,14)	$2(0) + 3(14) = 42$
E(1.3, 4.6)	$2(1.3) + 3(4.6) = 16.4$
B(6,0)	$2(6) + 3(0) = 12$

Result:

Hence $x_1 = 6$, $x_2 = 0$

Minimum Requirement of Vitamin is 12.

- 2 iv) A small workshop makes Tables and Chairs. Each table requires 4 hours of carpentry and 3 hours of painting. Each chair requires 3 hours of carpentry and 1 hour of painting. The workshop has at most 24 hours of carpentry and 9 hours of painting available per week. Profit is ₹ 60 per table and ₹ 30 per chair. Formulate and solve by using Graphical Method.

Solution:

Let x_1 = Number of tables

x_2 = Number of chairs

I. Formulation of LPP

Step 1: Objective of the problem is to Maximize profit.

Step 2: Decision Variables:

Let x_1 be the no. of tables produced

Let x_2 be the no. of chairs produced.

Decision Variables are x_1, x_2

Resources are Carpentry and Painting.

Step 3: Resource table

Machines	Tables (x_1)	Chairs (x_2)	Constraints
Carpentry	4	3	24
Painting	3	1	9
Profit	60	30	

Step 4: Formulation

Objective function : Maximize $Z = 60x_1 + 30x_2$

Subject to $4x_1 + 3x_2 \leq 24$

$3x_1 + x_2 \leq 9$

$x_1, x_2 \geq 0$

II. Solve by the LPP by Graphical Method

Step 1: Convert inequality into equality

$$4x_1 + 3x_2 = 24 \quad \text{--- (1)}$$

$$3x_1 + x_2 = 9 \quad \text{--- (2)}$$

Step 2: To plot the points

on equation (1)

$$4x_1 + 3x_2 = 24$$

$$\text{Put } x_1 = 0$$

$$0 + 3x_2 = 24$$

$$x_2 = 8 \quad \therefore \text{Point A is } (0, 8)$$

$$\text{Put } x_2 = 0$$

$$4x_1 + 0 = 24$$

$$x_1 = 6$$

$\therefore$ The point B is (6, 0)

On equation (2)

$$3x_1 + x_2 = 9$$

$$\text{Put } x_1 = 0$$

$$0 + x_2 = 9$$

$$x_2 = 9$$

$\therefore$ The point C is (0, 9)

$$\text{Put } x_2 = 0$$

$$3x_1 + 0 = 9$$

$$x_1 = 3$$

$\therefore$ The point D is (3, 0)

On solving Equation (1) & (2)

$$4x_1 + 3x_2 = 24$$

$$3x_1 + x_2 = 9$$

from eq (2)

$$x_2 = 9 - 3x_1$$

Subs in eq (1)

$$4x_1 + 3(9 - 3x_1) = 24$$

$$4x_1 + 27 - 9x_1 = 24$$

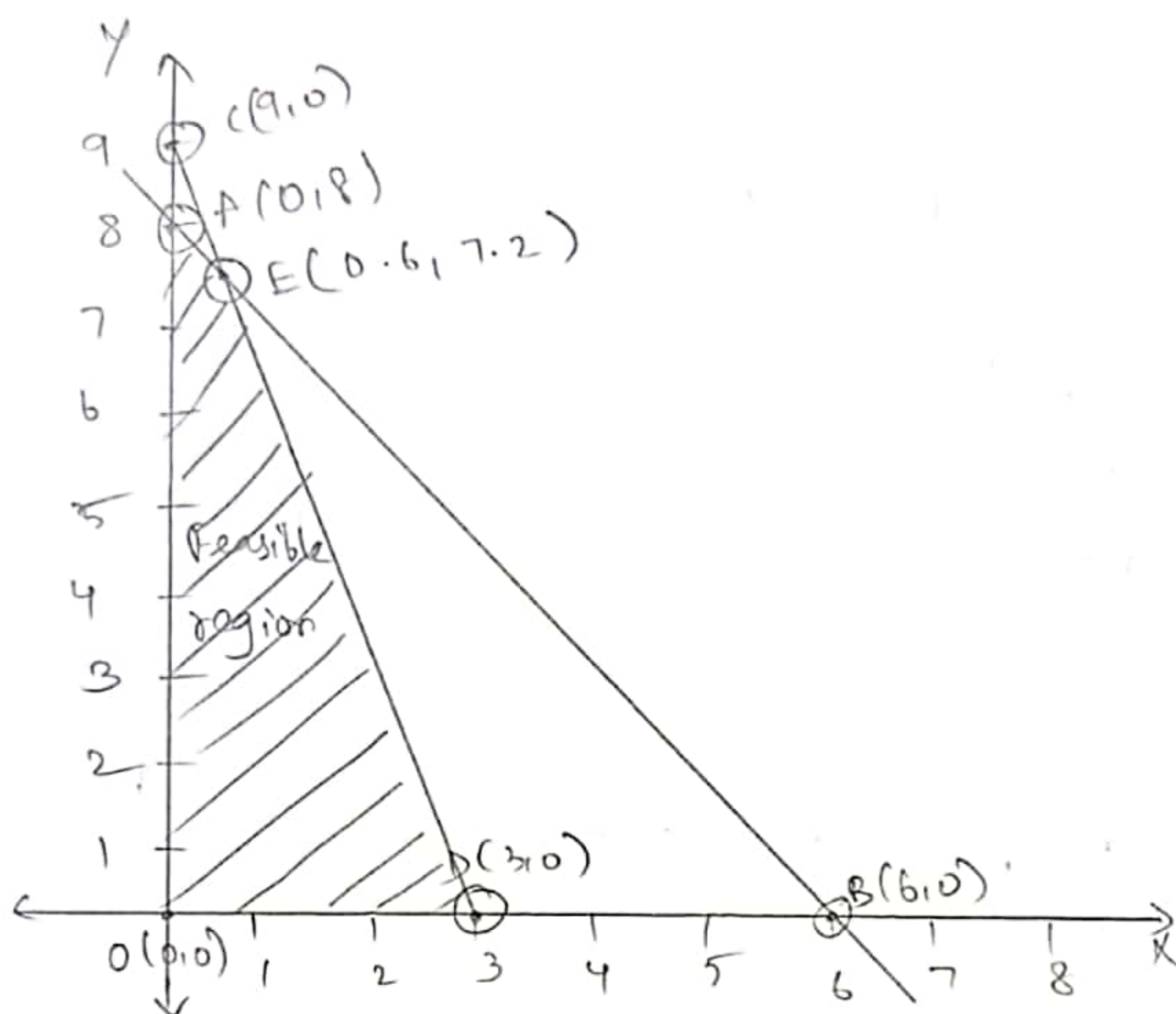
$$-5x_1 = -3$$

$$x_1 = \frac{3}{5} = 0.6$$

$$x_2 = 9 - 3(0.6) = 7.2$$

$\therefore$ The point of Intersection is E = (0.6, 7.2)

Step 3 : Graph



Step 4 : To find the optimal solution

Vertices	Objective function $\text{Max } Z = 60x_1 + 30x_2$
$O(0,0)$	$Z = 60(0) + 30(0) = 0$
$A(0,8)$	$Z = 60(0) + 30(8) = 240$
$B(3,0)$	$Z = 60(3) + 30(0) = 180$
$E(0.6, 7.2)$	$Z = 60(0.6) + 30(7.2) = 252$

Step 5 : Result

$$x_1 = 0.6 \rightarrow \text{Tables}, \quad x_2 = 7.2 \rightarrow \text{Chairs}$$

$$\therefore \text{Maximum profit} = \text{₹ } 252$$

3) f) Solve the following LPP using graphical method.

i) Maximize $Z = 3x_1 + 2x_2$
 subject to,
 $-2x_1 + x_2 \leq 1$
 $x_1 \leq 2$
 $x_1 + x_2 \leq 3$
 $x_1, x_2 \geq 0$

Solution:

Step 1: Convert from inequality to equality

$$-2x_1 + x_2 = 1 \quad \text{--- (1)}$$

$$x_1 = 2 \quad \text{--- (2)}$$

$$x_1 + x_2 = 3 \quad \text{--- (3)}$$

Step 2: plotting points

Or solving Eq (1),

$$\text{Put } x_1 = 0$$

$$0 + x_2 = 1$$

$$\therefore x_2 = 1$$

The point A is $(0, 1)$

Or solving eq (2)

$$\text{Put } x_1 = 0$$

$$\Rightarrow x_2 = 3$$

$\therefore$ The point C is $(0, 3)$

$$\text{Put } x_2 = 0$$

$$-2x_1 + 0 = 1$$

$$x_1 = -\frac{1}{2}$$

The point B is $(-\frac{1}{2}, 0)$

$$\text{Put } x_2 = 0,$$

$$\Rightarrow x_1 + 0 = 3$$

$$x_1 = 3$$

$\therefore$ The point D is
 $(3, 0)$

on solving ① & ②, we get

$$-2(2) + x_2 = 1$$

$$-4 + x_2 = 1$$

$$x_2 = 1 + 5$$

$$x_2 = 5$$

The point E is (2, 5)

on solving ① & ③,

$$x_1 = \frac{2}{3}; \quad x_2 = \frac{7}{3}$$

∴ The point A is $(\frac{2}{3}, \frac{7}{3})$
(0.66, 2.33)

On solving ② & ③

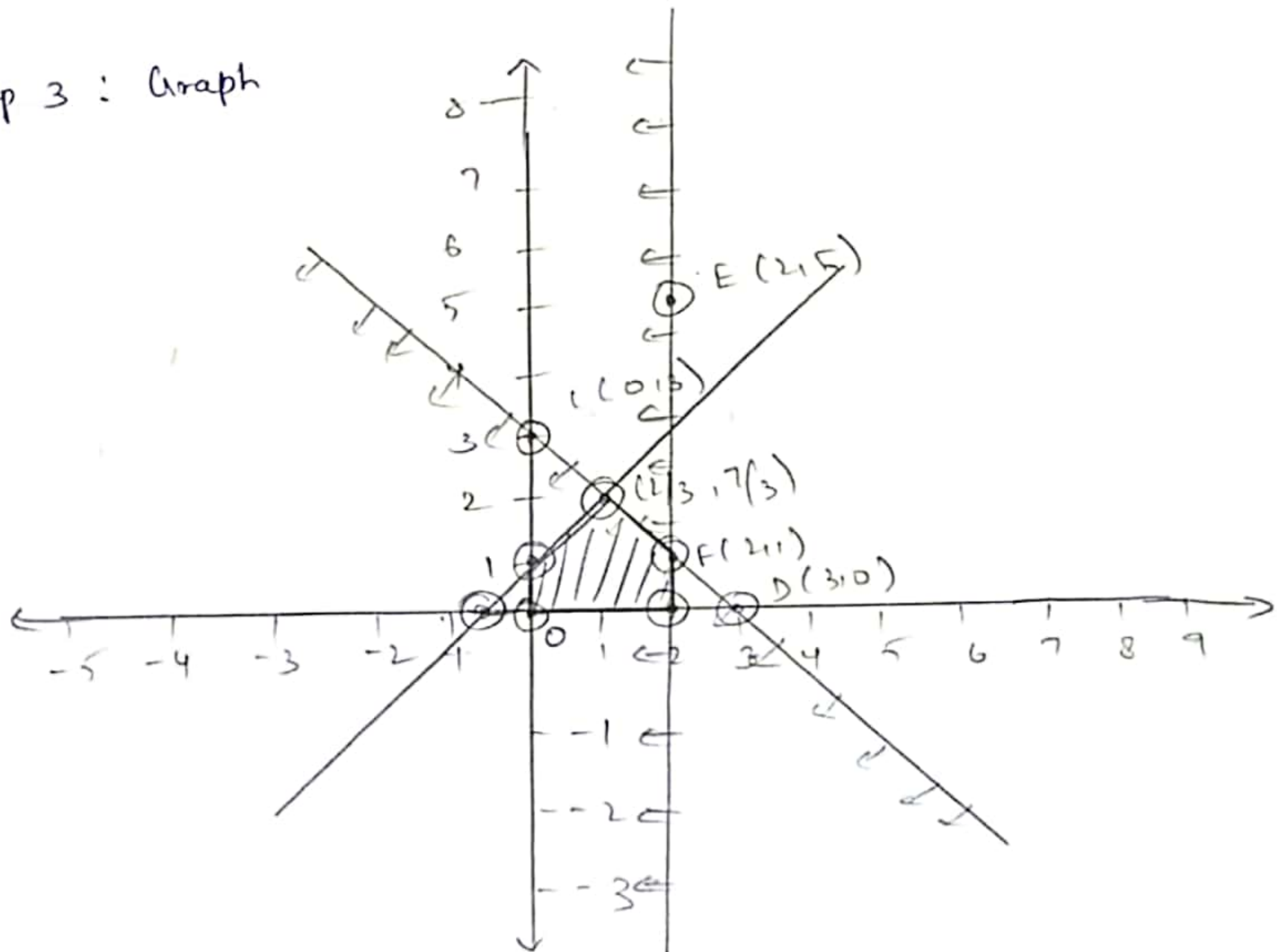
$$2 + x_2 = 3$$

$$x_2 = 3 - 2$$

$$x_2 = 1$$

∴ The point F is (2, 1)

Step 3 : Graph



Step 4 : To find optimal solution

Vertices	Objective function $\text{Max } Z = 3x_1 + 2x_2$
$O(0,0)$	$Z = 3(0) + 2(0) = 0$
$A(0,1)$	$Z = 3(0) + 2(1) = 2$
$C(2/3, 7/3)$	$Z = 3(2/3) + 2(7/3)$ $= 20/3 = 6.66$
$F(2,1)$	$Z = 3(2) + 2(1)$ $= 6 + 2 = 8$
$D(2,0)$	$Z = 3(2) + 2(0)$ $= 6$

Step 5 : Result

$$\therefore x_1 = 2, x_2 = 1$$

$$\text{Maximize } Z = 8.$$

3ii) Maximize $Z = x_1 - 2x_2$

$$\text{Subject to } -x_1 + x_2 \leq 1$$

$$6x_1 + 4x_2 \geq 24$$

$$0 \leq x_1 \leq 5 \text{ \& } 2 \leq x_2 \leq 4$$

Sol: Convert inequalities into equalities.

$$-x_1 + x_2 = 1 \text{ — (1)}$$

$$6x_1 + 4x_2 = 24 \text{ — (2)}$$

$$x_1 = 0 \text{ — (3)}$$

$$x_1 = 5 \text{ — (4)}$$

$$x_2 = 2 \text{ — (5)}$$

$$x_2 = 4 \text{ — (6)}$$

Step 2 : plotting the points

On solving eq (1),

$$\text{Put } x_1 = 0$$

$$0 + x_2 = 1$$

$$x_2 = 1$$

$\therefore$ The point A is (0, 1)

On solving eq (2)

$$\text{Put } x_1 = 0,$$

$$0 + 4x_2 = 24$$

$$4x_2 = 24$$

$$x_2 = 6$$

$\therefore$ The point C is (0, 6)

On solving (1) & (2)

$$x_1 = 2, x_2 = 3$$

$\therefore$ The point E is (2, 3)

On solving (4) & (5)

$$x_1 = 5, x_2 = 2$$

The point H is (5, 2)

$$\text{Put } x_2 = 0$$

$$-x_1 + 0 = -1$$

$$x_1 = -1$$

$\therefore$ The point B is (-1, 0)

$$\text{Put } x_2 = 0$$

$$6x_1 + 0 = 24$$

$$x_1 = 4$$

$\therefore$ The point D is (4, 0)

On solving (1) & (6)

The point D is (3, 4)

On solving (2) & (5)

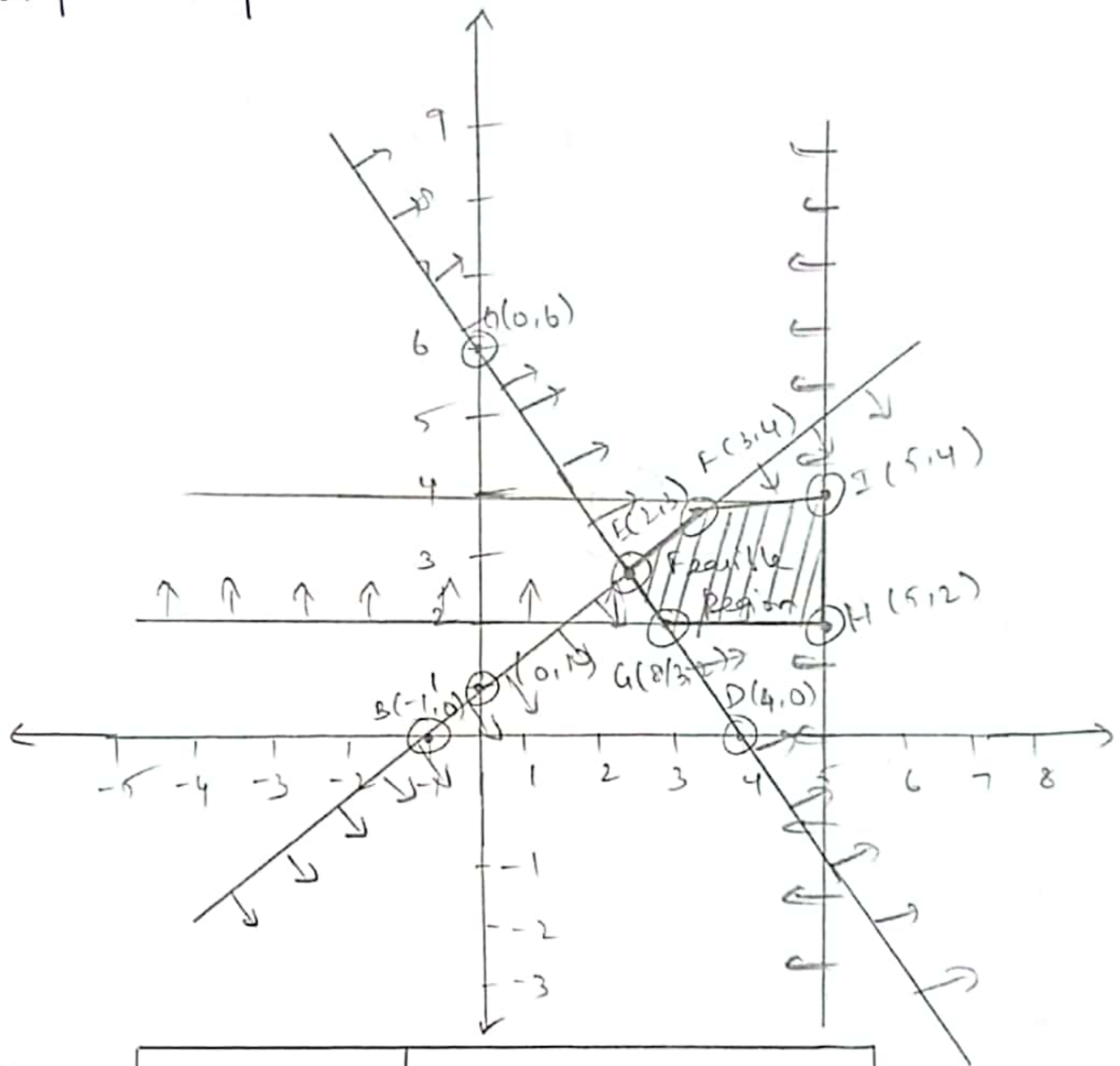
The point G is (8/3, 2)

On solving (4) & (6)

$$x_1 = 5, x_2 = 4$$

The point I is (5, 4)

Step 3: Graph



Step 4:

Vertices	Objective Function
$E(2,3)$	$\text{Max } z = x_1 - 2x_2$ $z = 2 - 2(3) = -4$
$F(3,4)$	$z = 3 - 2(4) = -5$
$I(5,4)$	$z = 5 - 2(4) = -3$
$H(5,2)$	$z = 5 - 2(2) = 1$
$G(8/3, 2)$	$z = \frac{8}{3} - 2(2) = -\frac{4}{3}$

Step 5:

Result $x_1 = 5, x_2 = 2$

Hence Maximum $z = 1$.

3) iii) Minimize $z = 45x_1 + 55x_2$

Subject to, $x_1 + 2x_2 \leq 30$

$2x_1 + 3x_2 \leq 80$

$x_1 - x_2 \geq 8$

$x_1, x_2 \geq 0$

Sol:

Step 1: Convert inequalities to equalities.

$x_1 + 2x_2 = 30$ — (1)

$2x_1 + 3x_2 = 80$ — (2)

$x_1 - x_2 = 8$ — (3)

Step 2: Plotting points

On solving eq (1),

put $x_1 = 0$

$0 + 2x_2 = 30$

$2x_2 = 30$

$x_2 = 15$

$\therefore$ The point A is (0, 15)

Put $x_2 = 0$,

$x_1 + 0 = 30$

$x_1 = 30$

$\therefore$ The point B is (30, 0)

On solving eq (2)

$$\text{Put } x_1 = 0$$

$$0 + 3x_2 = 80$$

$$x_2 = \frac{80}{3}$$

$\therefore$ The point C is $(0, 80/3)$

On solving eq (3)

$$x_1 = 0,$$

$$0 - x_2 = 8$$

$$x_2 = -8$$

$\therefore$ The point is $(0, -8)$

On solving (1) & (2)

$\therefore$ The point A is $(70, -20)$

On solving (1) & (3)

$\therefore$ The point I is $\left(\frac{46}{3}, \frac{22}{3}\right)$

15.3, 7.3

$$\text{Put } x_2 = 0$$

$$2x_1 + 3(0) = 80$$

$$2x_1 = 80$$

$$x_1 = 40$$

$\therefore$ The point D is $(40, 0)$

$$x_2 - 0 = 8$$

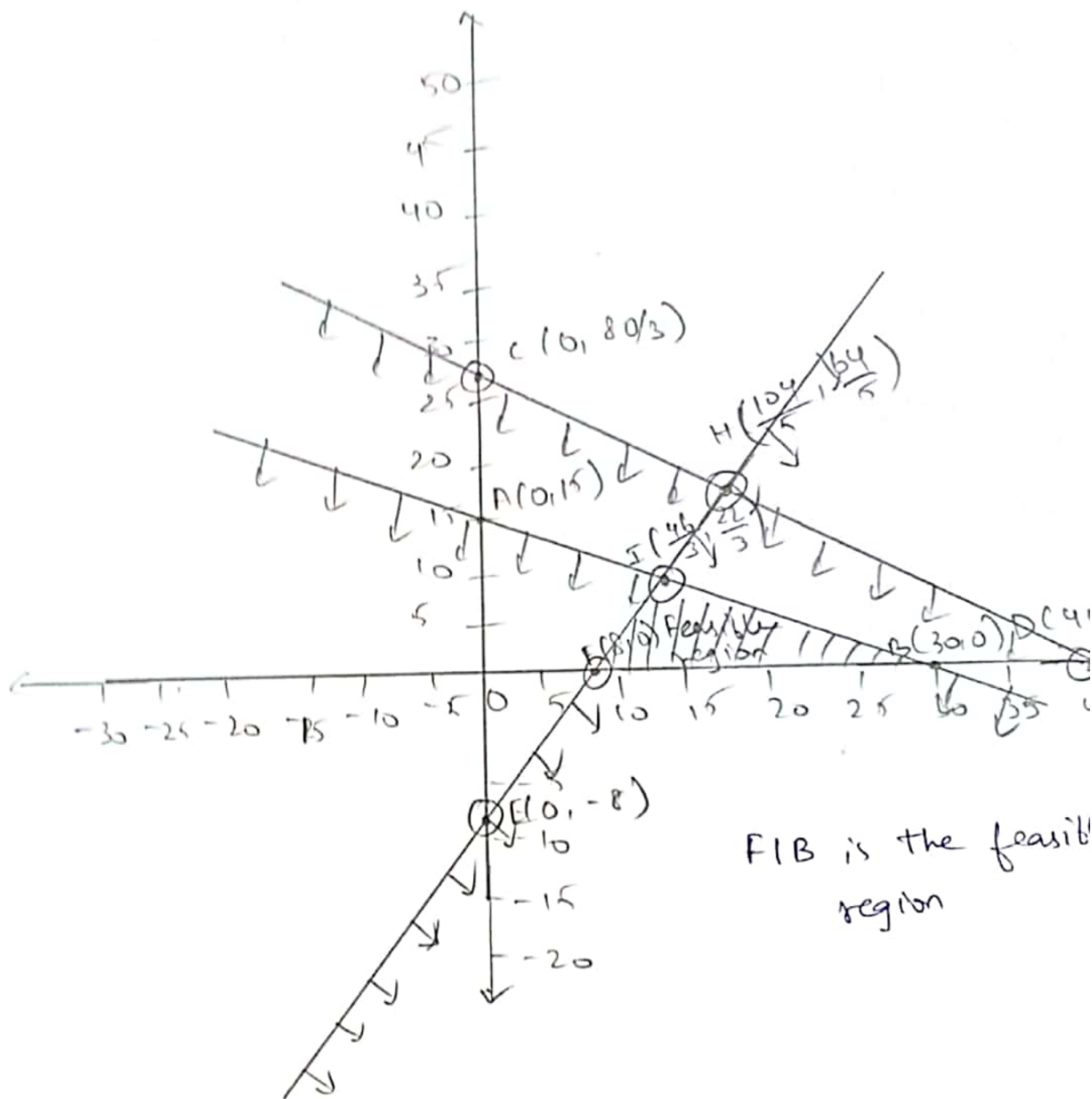
$$x_1 = 8$$

$\therefore$ The point F is $(8, 0)$

On solving (2) & (3)

$\therefore$ The point H $\left(\frac{104}{5}, \frac{64}{5}\right)$
20.8, 12.8

Step 3: Graph



FIB is the feasible region

Step 4: Optimal solution table

vertices	Objective function $\text{Min } Z = 45x_1 + 55x_2$
$F(8, 0)$	$Z = 45(8) + 55(0) = 360$
$E(\frac{46}{3}, \frac{22}{3})$	$Z = 45(\frac{46}{3}) + 55(\frac{22}{3}) = 1093.33$
$B(30, 0)$	$Z = 45(30) + 55(0) = 1350$

Step 5: Result

$$x_1 = 8, x_2 = 0$$

Hence, Minimize:

$$Z = 360$$

3) iv) Maximize $Z = 6x_1 + 8x_2$
 subject to, $5x_1 + 10x_2 \leq 60$
 $4x_1 + 4x_2 \leq 40$
 $x_1, x_2 \geq 0$

Solution:

Maximize $Z = 6x_1 + 8x_2$

Step 1: Convert inequality into equality

$$5x_1 + 10x_2 = 60 \quad \text{--- (1)}$$

$$4x_1 + 4x_2 = 40 \quad \text{--- (2)}$$

Step 2: plotting points

On solving equation (1)

$$5x_1 + 10x_2 = 60$$

$$\text{Put } x_1 = 0$$

$$10x_2 = 60$$

$$x_2 = 6$$

$\therefore$ The point A is (0, 6)

$$\text{Put } x_2 = 0$$

$$5x_1 = 60$$

$$x_1 = 12$$

$\therefore$ The point B is (12, 0)

On solving Equation ②

$$4x_1 + 4x_2 = 40$$

$$\text{Put } x_1 = 0$$

$$4x_2 = 40$$

$$x_2 = 10$$

∴ The point C is (0, 10)

$$\text{Put } x_2 = 0$$

$$4x_1 = 40$$

$$x_1 = 10$$

∴ The point D is (10, 0)

Finding the intersection point of ① and ②

$$5x_1 + 10x_2 = 60$$

$$4x_1 + 4x_2 = 40$$

Divide the ① by ⑤ :

$$x_1 + 2x_2 = 12$$

Divide the ② by 4 :

$$x_1 + x_2 = 10$$

$$x_1 + 2x_2 = 12$$

$$\begin{array}{r} (-) \quad (-) \quad (-) \\ x_1 + x_2 = 10 \end{array}$$

$$x_2 = 2$$

$$\therefore x_1 + 2 = 10$$

$$x_1 = 8$$

Hence the intersection point is E = (8, 2)

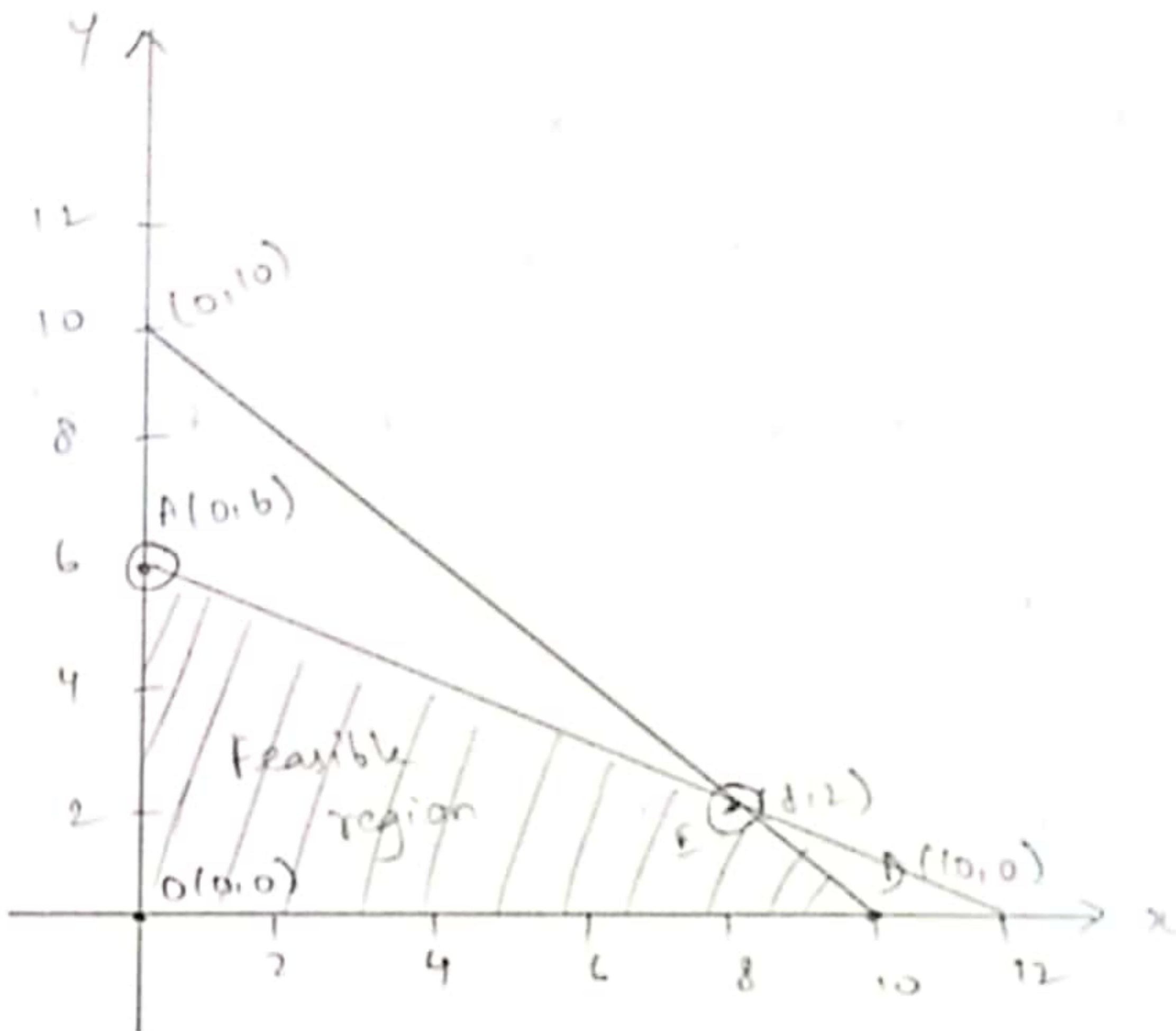
Step 3 : Graph

points : $O(0,0)$

$D(10,0)$

$E(8,2)$

$A(0,6)$



Step 4: Find the optimal solution

Vertex	Objective function $Z = 6x_1 + 8x_2$
O(0,0)	$Z = 6(0) + 8(0) = 0$
D(10,0)	$6(10) + 8(0) = 60$
E(8,2)	$6(8) + 8(2) = 64$
A(0,6)	$6(0) + 8(6) = 48$

The Maximum value of 64 is at E(8,2)

Step 5: Result

$$x_1 = 8, x_2 = 2$$

$$\therefore \text{Maximum } Z = 64$$

4) (i) Use Simplex Method to solve the following LPP.

$$\text{Minimize } z = 8x_1 - 2x_2$$

Subject to,

$$-4x_1 + 2x_2 \leq 1$$

$$5x_1 - 4x_2 \leq 3$$

$$x_1, x_2 \geq 0$$

Soln:

Step 1: The Given LPP can be reduced into

$$\begin{aligned} \text{Minimize } z &= -\text{Max}(z) \\ &= \text{Max } z^* \end{aligned}$$

$$\text{Maximize } z^* = -8x_1 + 2x_2$$

Subject to

$$-4x_1 + 2x_2 \leq 1$$

$$5x_1 - 4x_2 \leq 3$$

$$\text{where } x_1, x_2 \geq 0$$

By introducing non-negative slack variable s_1 & s_2 then the standard form of LPP becomes,

$$\text{Maximize } z^* = -8x_1 + 2x_2 + 0s_1 + 0s_2$$

Subject to

$$-4x_1 + 2x_2 + s_1 + 0s_2 = 1$$

$$5x_1 - 4x_2 + 0s_1 + s_2 = 3$$

$$\text{where } x_1, x_2, s_1 \text{ \& } s_2 \geq 0$$

$$n = 4$$

$$m = 2$$

$\Rightarrow n - m = 2$ variables become zero

$x_1 = x_2 = 0$: Non-basic variable

$S_1 = 1$; $S_2 = 3$: Basic variable

Step 2: Initial Iteration

C_B	Y_B	C_j	-8	2	0	0	$\theta = \min \left\{ \frac{x_{Bi}}{a_{ir}} : a_{ir} > 0 \right\}$
		x_B	x_1	x_2	S_1	S_2	
0	S_1	1	-4	2	1	0	$1/2 = 0.5 \rightarrow$
0	S_2	3	5	-4	0	1	-
Z_j		0	0	0	0	0	
$C_j - Z_j$		0	-8	2	0	0	

Here $C_j - Z_j > 0, \forall j$

The current IBFS is not optimal

x_2 enters the basis (Pivotal column)

S_1 leaves the basis (Pivotal row)

Pivotal element is 2

First Iteration:

C_B	Y_B	x_j	-8	2	0	0
		x_B	x_1	x_2	s_1	s_2
2	x_2	$1/2$	$-4/2$	1	$1/2$	0
0	s_2	5	-3	0	2	1
Z_j		$1+0$ $=1$	$-4+0$ $=-4$	$2+0$ $=2$	$1+0$ $=1$	$0+0$ $=0$
$C_j - Z_j$		1	-4	0	-1	0

Here $C_j - Z_j \leq 0 \forall j$

The current IBFS is optimal

$$x_1 = 0; x_2 = 1/2$$

$$\therefore \text{Maximize } z^* = 1$$

$$\text{Minimize } z = [-\text{Max } z^*] = -1$$

4)(i) Use Simplex Method to solve the following LPP.

$$\text{Maximize } Z = 10x_1 + 15x_2 + 20x_3$$

Subject to,

$$2x_1 + 4x_2 + 6x_3 \leq 24$$

$$3x_1 + 9x_2 + 6x_3 \leq 30$$

$$x_1, x_2, x_3 \geq 0$$

Soln:

Step 1: Rewriting the Constraints

$$\text{Max } Z = 10x_1 + 15x_2 + 20x_3$$

Subject to

$$x_1 + 2x_2 + 3x_3 \leq 12$$

$$x_1 + 3x_2 + 2x_3 \leq 10$$

$$\text{Where } x_1, x_2, x_3 \geq 0$$

Step 2: By introducing the non-negative slack variables s_1 & s_2 in the form of the LPP becomes Maximize

$$\text{Maximize } Z = 10x_1 + 15x_2 + 20x_3 + 0s_1 + 0s_2$$

Subject to,

$$x_1 + 2x_2 + 3x_3 + s_1 + 0s_2 = 12$$

$$x_1 + 3x_2 + 2x_3 + 0s_1 + s_2 = 10$$

$$\text{Where } x_1, x_2, x_3, s_1 \text{ \& } s_2 \geq 0$$

Here $n=5$ (no. of variables)

$m=2$ (no. of equations)

$\therefore n-m=3$ variables to zero (first 3 variables is 0)

$x_1 = x_2 = x_3 = 0$ (Non-Basic variables)

[Substitute in ①, ②]

$S_1 = 12$; $S_2 = 10$ (Basic variables)

Initial Iteration:

C_B	X_B	C_i	10	15	20	0	0	$\theta = \min \left\{ \frac{x_{0i}}{a_{ij}} \right\}$ $a_{ij} > 0$
		X_B	x_1	x_2	x_3	S_1	S_2	
0	S_1	12	1	2	③	1	0	$\frac{12}{3} = ④ \rightarrow$
0	S_2	10	1	3	②	0	1	$\frac{10}{2} = 5$
Z_j		0	0	0	0	0	0	
$C_j - Z_j$		0	10	15	20	0	0	

where $C_j - Z_j \geq 0$, The current IBFS is not optimal

x_3 enters the Basis

S_1 leaves the Basis

$\therefore$ Pivotal element is 3

First Iteration:

C_B	Y_B	C_j	10	15	20	0	0	$\theta = \min \left\{ \frac{x_{B_i}}{a_{i3}}, a_{i3} > 0 \right\}$
		x_B	x_1	x_2	x_3	s_1	s_2	
20	x_3	$\frac{12}{3} = 4$	$\frac{1}{3}$	$\frac{2}{3}$	1	$\frac{1}{3}$	0	$\frac{4}{1/3} = 12$
0	s_2	2	$\frac{1}{3}^*$	$5/3$	0	$-2/3$	1	$2/(1/3) = 6 \rightarrow$
Z_j		$(20 \times 4) + (0 \times 2) = 80$	$20/3$	$40/3$	20	$20/3$	0	
$C_j - Z_j$		80	$10/3 = 3.3$	$5/3 = 1.67$	0	$-20/3$	0	

Here $C_j - Z_j \geq 0$, The current IBFS is not optimal.

So x_1 enters the basis; s_2 leaves the basis

$\therefore$ Pivotal element is $1/3$

Second Iteration

C_B	Y_B	C_j	10	15	20	0	0
		X_B	x_1	x_2	x_3	s_1	s_2
20	(x_3)	2	0	-1	1	1	-1
10	(x_1)	$2/1/3 = 6$	1	$5/3/1/3 = 5$	0	$-2/3/1/3 = -2$	$1/3 = 3$
Z_j		$40+60 = 100$	$0+10 = 10$	$-20+50 = 30$	$20+0 = 20$	$20-20 = 0$	$-20+30 = 10$
$C_j - Z_j$		100	0	-15	0	0	-10

Since $C_j - Z_j \leq 0 \forall j$

The current IBFS is ~~optional~~ optimal

Result:

Hence, max $x_1 = 6$

$x_2 = 0$

$x_3 = 2$

$Z = 100$

Verification

$\therefore \text{max } Z = 10x_1 + 15x_2 + 20x_3$

$Z = 10(6) + 15(0) + 20(2)$

$= 60 + 0 + 40$

$= 100$

4)(iii) Use Simplex Method to solve the following LPP.

$$\text{Maximize } Z = 6x_1 + 8x_2$$

S.T.

$$5x_1 + 10x_2 \leq 60$$

$$4x_1 + 4x_2 \leq 40$$

$$x_1, x_2 \geq 0$$

Soln:

Step 1: Rewriting the constraints

$$\text{Maximize } Z = 6x_1 + 8x_2$$

S.T.

$$x_1 + 2x_2 \leq 12$$

$$x_1 + x_2 \leq 10$$

$$\text{Where } x_1, x_2 \geq 0$$

Step 2:

By introducing the non-negative slack variable s_1 & s_2 into standard form of the LPP becomes.

$$\text{Maximize } Z = 6x_1 + 8x_2 + 0s_1 + 0s_2$$

S.T.

$$x_1 + 2x_2 + s_1 + 0s_2 = 12$$

$$x_1 + x_2 + 0s_1 + s_2 = 10$$

$$\text{Where } x_1, x_2, s_1, s_2 \geq 0$$

Here no. of Equations (m) = 2

No. of variables (n) = 4

$$n - m = 4 - 2 = 2$$

2 Variables to zero

Non-Basic Variables $x_1 = 0, x_2 = 0$

$S_1 = 12$ } Basic Variables
 $S_2 = 10$

Initial Iteration:

C_B	Y_B	C_j	6	8	0	8	$\theta = \min \left\{ \frac{x_{Bj}}{a_{ij}}, a_{ir} > 0 \right\}$
		Y_B	x_1	x_2	S_1	S_2	
0	S_1	12	1	2	1	0	$\frac{12}{2} = 6 \rightarrow$
0	S_2	10	1	1	0	1	$\frac{10}{1} = 10$
Z_j		(const) (const)=0	0	0	0	0	
$C_j - Z_j$		0	6	8	0	0	

Since $C_j - Z_j > 0$, The current BFS is not optimal.

So x_2 enters the Basis
 S_1 leaves the Basis

$\therefore$ Pivotal element is = 2

First Iteration:

C_B	Y_B	C_j	6	8	0	0	$\theta = \min \left\{ \frac{x_{B_i}}{a_{ir}}, a_{ir} > 0 \right\}$
		Y_B	x_1	x_2	s_1	s_2	
8	x_2	$\frac{12}{2} = 6$	$\frac{1}{2}$	1	$\frac{1}{2}$	0	$6 / \frac{1}{2} \Rightarrow 6 \times 2 = 12$
0	s_2	$\frac{8}{2} = 4$	$\frac{1}{2}$	0	$-\frac{1}{2}$	1	$4 / \frac{1}{2} \Rightarrow 4 \times 2 = 8 \rightarrow$
Z_j		$= 48$	4	8	4	0	
$C_j - Z_j$		48	2	0	-4	0	

So x_1 enters the basis, s_2 leaves the basis
 $\therefore$ Pivotal element is $\frac{1}{2}$

Since $C_j - Z_j > 0$ for all j . The current IBFS is not optimal

Second Iteration:

C_B	Y_B	C_j	6	8	0	0
		x_B	x_1	x_2	s_1	s_2
8	x_2	$1/2 = 2$	0	$\frac{1/2}{1/2} = 1$	1	$\frac{-1/2}{1/2} = -1$
6	x_1	$4/1 = 4$	1	$0/1 = 0$	$\frac{-1/2}{1/2} = -1$	$1/1 = 1$
Z_j		64	6	8	2	-8
$C_j - Z_j$		64	0	0	-2	-4

Since $C_j - Z_j \leq 0$ for all j , the current BFS is optimal

Result:

$$x_1 = 8, x_2 = 2$$

$$\begin{aligned}
 \text{Maximize } Z &= 6x_1 + 8x_2 \\
 &= 6(8) + 8(2) \\
 &= 64
 \end{aligned}$$

4) iv) Use Simplex Method to solve the following LPP.

$$\text{Maximize } Z = 3x_1 + 5x_2$$

S.T.

$$x_1 + 2x_2 \leq 8$$

$$3x_1 + 2x_2 \leq 12$$

$$x_1, x_2 \geq 0$$

Soln:

Step 1: Rewriting the constraints

$$\text{Maximize } Z = 3x_1 + 5x_2$$

S.T.

$$x_1 + 2x_2 \leq 8$$

$$3x_1 + 2x_2 \leq 12$$

$$\text{where } x_1, x_2 \geq 0$$

Step 2:

By introducing the non-negative slack variables s_1 & s_2 into the standard form of the LPP becomes.

$$\text{Maximize } Z = 3x_1 + 5x_2 + 0s_1 + 0s_2$$

S.T.

$$~~x_1 + 2x_2 + 0s_1~~$$

$$x_1 + 2x_2 + s_1 + 0s_2 = 8$$

$$3x_1 + 2x_2 + 0s_1 + s_2 = 12$$

$$\text{where } x_1, x_2, s_1 \text{ \& } s_2 \geq 0$$

Here $m = 2$

$n = 4$

$n - m = 2$ Variables to zero

Non - Basic Variables $x_1 = 0, x_2 = 0$

$S_1 = 8$
 $S_2 = 12$ } Basic variables

Initial Iteration:

C_B	Y_B	C_j	3	5	0	0	θ
		X_B	x_1	x_2	S_1	S_2	
0	S_1	8	1	2	1	0	$\frac{8}{2} = 4 \rightarrow$
0	S_2	12	3	2	0	1	$\frac{12}{2} = 6$
Z_j		0	0	0	0	0	
$C_j - Z_j$		0	3	5	0	0	

Entering variable: x_2

Leaving variable: S_1

Pivot element: 2

Since $C_j - Z_j > 0$ for some j , the current solution is not optimal.

First Iteration:

C_B	X_B	C_j	3	5	0	0	θ
		x_B	x_1	x_2	s_1	s_2	
5	x_2	4	$1/2$	1	$1/2$	0	$4/1/2 = 8$
0	s_2	4	2	0	-1	1	$4/2 = 2 \rightarrow$
Z_j		20	$5/2$	5	$5/2$	0	
$C_j - Z_j$		20	$1/2$	0	$-5/2$	0	

Since $C_j - Z_j > 0$ for x_1 , the current solution is not optimal.

Entering variable: x_1

Leaving variable: s_2

Pivot element: 2

5) i) Use Big-M method to solve the following LPP.

$$\text{Minimize } Z = 2x_1 + 3x_2$$

S.T.

$$x_1 + x_2 \geq 6$$

$$7x_1 + x_2 \geq 14$$

$$x_1, x_2 \geq 0$$

Soln:

By introducing surplus variable

The standard form of the LPP becomes

$$\text{Maximize } Z^* = -2x_1 - 3x_2 + 0s_1 + 0s_2 - MR_1 - HR_2$$

S.T.

$$x_1 + x_2 - s_1 + 0s_2 + R_1 = 6$$

$$7x_1 + x_2 + 0s_1 + s_2 + R_2 = 14$$

Surplus Variables : s_1 & s_2

Artificial Variables : R_1 & R_2

$$n = 6, m = 2$$

$$\therefore n - m = 6 - 2$$

= 4 Variables becomes zero

$$x_1 = x_2 = s_1 = s_2 = 0 \text{ (Non-Basic variables)}$$

$$R_1 = 6; R_2 = 14$$

Second Iteration:

C_B	Y_B	C_j	3	5	0	0
		Y_B	x_1	x_2	S_1	S_2
5	x_2	3	0	1	$3/4$	$-1/4$
3	x_1	2	1	0	$-1/2$	$1/2$
	Z_j	21	3	5	$9/4$	$1/4$
	$C_j - Z_j$	21	0	0	$-9/4$	$-1/4$

Since $C_j - Z_j \leq 0$ for all j , the current solution is optimal.

Result:

$$x_1 = 2$$

$$x_2 = 3$$

$$\text{Maximum } z = 21$$

Initial Iteration

C_B	Y_B	C_j	-2	-3	0	0	-M	-M	$\theta = \min$
		x_B	x_1	x_2	s_1	s_2	R_1	R_2	
-M	R_1	6	1	1	-1	0	1	0	$6/7 = 6$
-M	R_2	14	7	1	0	-1	0	1	$14/7 = 2 \rightarrow$
Z_j		-20M	-8M	-2M	M	M	-M	-M	
$C_j - Z_j$		-20M	$8M$ -2	-3+2M	-M	-M	0	0	

Here $Z_j - C_j \geq 0 \forall j$

The current IBFS is not ~~optional~~ optimal

So, x_1 enter the Basis

R_2 leaves the Basis & Pivotal element is 7

First Iteration:

C_B	Y_B	C_j	-2	-3	0	0	-M	$\theta = \min$
		X_B	x_1	x_2	s_1	s_2	R_1	
-M	R_1	4	0	$6/7$	-1	$1/7$	1	$\frac{4}{6/7} = \frac{28}{6} = 4.67 \rightarrow$
-2	x_1	2	1	$1/7$	0	$-1/7$	0	$2/1/7 = 14$
Z_j		$-4M-4$	-2	$-\frac{6M-2}{7}$	M	$-\frac{M}{7} + \frac{2}{7}$	-M	
$C_j - Z_j$		$-4M-4$	0	$\frac{6M-19}{7}$	-M	$\frac{M-2}{7}$	0	

Here $C_j - Z_j \geq 0 \forall j$

The current BFS is not optimal

So, x_2 is entering the basis

R_1 leaves the basis

$\therefore$ The pivotal element is $6/7$

Second Iteration:

C_B	Y_B	C_j	-2	-3	0	0	$\theta = \min$
		X_B	x_1	x_2	s_1	s_2	
-3	x_2	$28/6$	0	1	$-7/6$	$1/6$	$\frac{28/6}{1/6} = 28 \rightarrow$
-2	x_1	$8/6$	1	0	$1/6$	$-1/6$	-
Z_j		$-100/6$	-2	-3	$19/6$	$-1/6$	
$C_j - Z_j$		$-100/6$	0	0	$-19/6$	$1/6$	$\uparrow$

Here $C_j - Z_j \geq 0 \forall j$, The current BFS is not optimal.

So, s_2 enters the basis

x_2 leaves the basis.

Pivotal element is $1/6$

Third Iteration:

C_B	Y_B	C_j	-2	-3	0	0
		X_B	x_1	x_2	s_1	s_2
0	s_2	28	0	6	-7	1
-2	x_1	6	1	1	-1	0
Z_j		-12	-2	-2	2	0
$C_j - Z_j$		-12	0	-1	-2	0

Here $C_j - Z_j \leq 0$,

Then the current BFS is optimal

$$x_1 = 6, x_2 = 0$$

Thus the Maximize $z^* = -12$

$$\Delta \text{ minimize} = -\text{Max } z^*$$

$$= -(-12)$$

$$= 12$$

5) ii) Use Big M-method to solve the following LPP.

$$\text{Maximize } z = 20x_1 + 10x_2 + 15x_3$$

S.T.

$$8x_1 + 6x_2 + 2x_3 \leq 60$$

$$5x_1 + x_2 + 6x_3 \geq 40$$

$$2x_1 + 6x_2 + 3x_3 \leq 30$$

$$x_1, x_2, x_3 \geq 0$$

Soln:

Step 1: By introducing non-negative slack, surplus and artificial variable into the standard form of LPP becomes.

$$\text{Maximize } z = 20x_1 + 10x_2 + 15x_3 + 0s_1 + 0s_2 + 0s_3 - MR_1$$

S.T.

$$8x_1 + 6x_2 + 2x_3 + s_1 + 0s_2 + 0s_3 + 0R_1 = 60$$

$$5x_1 + x_2 + 6x_3 + 0s_1 - s_2 + 0s_3 + R_1 = 40$$

$$2x_1 + 6x_2 + 3x_3 + 0s_1 + 0s_2 + s_3 + 0R_1 = 30$$

$$\text{where } x_1, x_2, x_3 \geq 0$$

Slack variables: $s_1, s_3 \geq 0$

Surplus variables: $s_2 \geq 0$

Artificial variables: $R_1 \geq 0$

$$n = 7$$

$$m = 3$$

$$n - m = 7 - 3 = 4$$

$$x_1 = x_2 = x_3 = s_2 = 0 \text{ (Non-Basic variables)}$$

$$\begin{matrix} s_1 = 60 \\ R_1 = 40 \end{matrix} \quad s_3 = 30 \quad \left. \vphantom{\begin{matrix} s_1 = 60 \\ R_1 = 40 \end{matrix}} \right\} \text{Basic Variables}$$

Initial Iteration:

C_B	Y_B	C_j	20	10	15	0	0	0	-M	0
		X_B	x_1	x_2	x_3	s_1	s_2	s_3	R_1	
0	s_1	60	8	6	2	1	0	0	0	
-M	R_1	40	5	1	6	0	-1	0	1	
0	s_3	30	2	6	3	0	0	1	0	
	Z_j	-40M	-5M	-M	-6M	0	M	0	-M	
	$C_j - Z_j$	-40M	20+5M	10+M	6M+15	0	-M	0	0	

Here $C_j - Z_j \geq 0$ is $\forall_j$,

The current IBFS is not optimal

So x_3 enters the basis

R_1 leaves the basis & pivotal element is 6

First Iteration:

C_B	Y_B	C_j	20	10	15	0	0	0	$\theta =$
		X_B	x_1	x_2	x_3	s_1	s_2	s_3	
0	s_1	$\frac{140}{3}$	$19/3$	$17/3$	0	1	$1/3$	0	$140 / \frac{19}{3} = 21.05$
15	x_3	$40/6$	$5/6$	$1/6$	1	0	$-1/6$	0	
0	s_3	10	$-1/2$	$11/2$	0	0	$1/2$	1	
					15	0	$-5/2$	0	
Z_j		100	$25/2$	$5/2$					
$C_j - Z_j$		100	$15/2$	$15/2$	0	0	$5/2$	0	

Hence $C_j - Z_j \geq 0 \forall j$, The current ZBFS is not optimal. So x_1 enters the basis and s_1 exits the basis. $\therefore$ Pivotal element is $19/3$.

Second Iteration:

C_B	X_B	C_j	20	10	15	0	0	θ
		x_B	x_1	x_2	x_3	s_2	s_3	
20	x_1	$140/19$	1	$17/19$	0	$1/19$	0	$140/19 \cdot 1/19 = 140$
15	x_3	$10/19$	0	$-11/19$	1	$-14/19$	0	-
0	s_3	$260/19$	0	$173/19$	0	$10/19$	1	$\frac{260}{19} / \frac{10}{19} = 26 \rightarrow$
Z_j		$2950/19$	20	$175/19$	15	$-40/19$	0	
$C_j - Z_j$		$2950/19$	0	$15/19$	0	$40/19$	0	

Here $C_j - Z_j \geq 0$, $\therefore$ The current BFS is not optimal

So s_2 enters the base

s_3 leaves the base

Pivotal element is $10/19$

Third Iteration:

C_B	Y_B	C_j	20	10	15	0
		Y_B	x_1	x_2	x_3	s_2
20	x_1	6	1	$3/10$	0	0
15	x_3	6	0	$9/5$	1	0
0	s_2	26	0	$113/10$	0	1
Z_j		210	20	33	15	0
$C_j - Z_j$		210	0	-23	0	0

$C_j - Z_j \leq 0$, The current IBFS is optimal

$$\text{Max } x_1 = 6 \quad x_3 = 6$$

$$x_2 = 6$$

$$\text{Max } Z = 210.$$

r)iii) Use Big M-method to solve the following LPD.

$$\text{Minimize } z = 12x_1 + 18x_2 + 15x_3$$

s.t.

$$4x_1 + 8x_2 + 6x_3 \geq 64$$

$$3x_1 + 6x_2 + 12x_3 \geq 96$$

$$x_1, x_2, x_3 \geq 0$$

Soln:

Dividing the constraints inequalities by common factors simplifies the algebra:

To apply the Big M-method, convert the objective to Maximization ($z^* = -z$) and subtract surplus variables (s_1, s_2) while adding artificial variables (A_1, A_2) with a large penalty coefficient $M > 0$:

$$\text{Maximize } z^* = -12x_1 - 18x_2 - 15x_3 + 0s_1 + 0s_2 - MA_1 - MA_2$$

s.t.

$$(\div) 2 \quad 2x_1 + 4x_2 + 3x_3 - s_1 + A_1 = 32$$

$$(\div) 3 \quad x_1 + 2x_2 + 4x_3 - s_2 + A_2 = 32$$

$$\text{where } x_1, x_2, x_3, s_1, s_2, A_1, A_2 \geq 0$$

Initial Iteration:

C_B	Y_B	C_j	-12	-18	-15	0	0	-M	-M	θ
		x_B	x_1	x_2	x_3	s_1	s_2	A_1	A_2	
-M	A_1	32	2	4	3	0-1	0	1	0	$3\frac{1}{4}=8$
-M	A_2	32	1	2	4	0	-1	0	1	$3\frac{1}{2}=16$
Z_j		-64M	-3M	-6M	-7M	M	M	-M	-M	
$C_j - Z_j$		-64M	3M-12	6M-18	7M-15	-M	-M	0	0	

Entering variable: x_2 Leaving variable: A_1

Pivot element: 4

First Iteration:

C_B	Y_B	C_j	-12	-18	-15	0	0	-M	θ
		x_B	x_1	x_2	x_3	s_1	s_2	A_2	
-18	x_2	8	$1/2$	1	$3/4$	$-1/4$	0	0	$8/(3/4) = \frac{32}{3}$
-M	A_2	16	0	0	$5/2$	$1/2$	-1	1	$\frac{16}{5/2} = \frac{32}{5}$
Z_j		-144	-9	-18	$-\frac{27}{2} - \frac{5}{2}M$	$\frac{9}{2} - \frac{1}{2}M$	M	-M	
$C_j - Z_j$		-44	-3	0	$\frac{5}{2}M - \frac{27}{2}$	$\frac{1}{2}M - \frac{9}{2}$	-M	0	

Second Iteration:

C_B	Y_B	C_j	-12	-18	-15	0	0	0
		Y_B	x_1	x_2	x_3	s_1	s_2	
-18	x_2	$16/5$	$1/2$	1	0	$-2/5$	$3/10$	$\frac{16/5}{1/2} = \frac{32}{5}$
-15	x_3	$32/5$	0	0	1	$1/5$	$-2/5$	
Z_j		$-768/5$	-9	-18	-15	$2/5$	$3/5$	
$C_j - Z_j$		$-768/5$	-3	0	0	$-2/5$	$-3/5$	

Since $C_j - Z_j \leq 0$ for all variables, the current solution is optimal.

Result:

$$x_1 = 0$$

$$x_2 = \frac{16}{5} = 3.2$$

$$x_3 = \frac{32}{5} = 6.4$$

$$\text{Minimum } Z = -Z^* = 153.6$$

6)i) Use two-phase method to solve the following LPP

$$\text{Minimize } Z = 12x_1 + 18x_2 + 15x_3$$

S.T.

$$4x_1 + 8x_2 + 6x_3 \geq 64$$

$$3x_1 + 6x_2 + 12x_3 \geq 96$$

$$x_1, x_2, x_3 \geq 0$$

Soln:

Step 1: Given LPP is rewritten as,

$$\begin{aligned} \text{Minimize } Z^* &= -\text{Max}(-Z) \\ &= \text{Max } Z^* \end{aligned}$$

$$\text{Max } Z^* = -12x_1 - 18x_2 - 15x_3$$

S.T.

$$8x_1 + 4x_2 + 3x_3 \geq 32$$

$$x_1 + 2x_2 + 4x_3 \geq 32$$

$$\text{where } x_1, x_2, x_3 \geq 0$$

Step 2: By introducing non-negative surplus and artificial variable into the standard form of the LPP becomes,

$$\text{Max } Z^* = -12x_1 - 18x_2 - 15x_3$$

s.t.

$$2x_1 + 4x_2 + 3x_3 - s_1 + 0s_2 + R_1 + 0R_2 = 32$$

$$x_1 + 2x_2 + 4x_3 + 0s_1 - s_2 + R_2 + 0R_1 = 32$$

$$\text{Where } x_1, x_2, x_3, s_1, s_2, R_1, R_2 \geq 0$$

$$n = 7$$

$$m = 2$$

$\therefore n - m = 5$ variables becomes zero

$$x_1 = x_2 = x_3 = s_1 = s_2 = 0 \text{ (Non basic variables)}$$

$$\left. \begin{array}{l} R_1 = 32 \\ R_2 = 32 \end{array} \right\} \text{Basic variables.}$$

Phase - I

The objective function of the auxiliary LP becomes

$$\text{Max } Z^* = -R_1 - R_2$$

Initial Simplex Table

C_B	x_B	C_j	0	0	0	0	0	-1	-1	0
		x_0	x_1	x_2	x_3	s_1	s_2	r_1	r_2	
-1	R_1	32	2	4	3	-1	0	1	0	$\frac{32}{3} = 10.67$
-1	R_2	32	1	2	4	0	-1	0	1	$\frac{32}{4} = 8$
	Z_j	-64	-3	-6	-7	1	1	-1	-1	
	$C_j - Z_j$	-64	3	6	-7	-1	-1	0	0	

Here $C_j - Z_j \geq 0$ $\forall j$ the current IBF is not optimal

So, x_3 enters the basis

R_2 leaves the basis

Pivotal element 4

First Iteration:

C_B	Y_B	C_j	0	0	0	0	0	-1	θ
		x_0	x_1	x_2	x_3	s_1	s_2	R_1	
-1	R_1	8	$5/4$	$5/2$	0	-1	$3/4$	1	$8/5/2 = 16/5 \rightarrow$
0	x_3	8	$1/4$	$1/2$	1	0	$-1/4$	0	$8/1/2 = 16$
Z_j		-8	$-5/4$	$-5/2$	0	1	$-3/4$	-1	
$C_j - Z_j$		-8	$5/4$	$5/2$	0	-1	$3/4$	0	

Here $C_j - Z_j \geq 0 \forall j$, The current TBFS is not optimal

So, x_2 enter the basis

R_1 leaves the basis

Pivotal element is $5/2$

Second Iteration:

C_B	Y_B	C_j	0	0	0	0	0
		X_B	x_1	x_2	x_3	s_1	s_2
0	x_2	$16/5$	$1/2$	1	0	$-2/5$	$3/10$
0	x_3	$32/5$	0	0	1	$1/5$	$-2/5$
Z_j		0	0	0	0	0	0
$C_j - Z_j$		0	0	0	0	0	0

$$\text{Here } C_j - Z_j \leq 0 \quad \forall j$$

The current IBFS is optimal and feasible

So, further, the set of basic variables in the optimal of phase - I does not contain Artificial variable so the given problem is feasible.

Phase-II

Here, we consider the actual cost associated with the original variables, then the new objective function becomes.

$$\text{Max } Z = 12x_1 + 18x_2 + 15x_3 + 0s_1 + 0s_2$$

The IBF for this phase is one obtained at the end of the phase I

The Iterative Simplex table for this phase are.

$$\text{Max } Z = -12x_1 - 18x_2 - 15x_3 + 0s_1 + 0s_2$$

C_B	X_B	C_j	-12	-18	-15	0	0
		X_B	x_1	x_2	x_3	s_1	s_2
-18	x_2	$16/5$	$1/2$	1	0	$-2/5$	$3/10$
-12	x_3	$32/5$	0	0	1	$1/5$	$-2/5$
Z_j		$-288 + 480$ $= 768/5$	-9	-18	-15	$21/5$	$6/10$
$C_j - Z_j$		$-768/5$	-3	0	0	$-21/5$	$-6/10$

Here $C_j - Z_j \leq 0 \forall j$

The current BFS is optimal and feasible

$$\therefore x_1 = 0$$

$$x_2 = 16/5$$

$$x_3 = 32/5$$

$$\text{Maximize } z^1 = -768/5$$

$$\begin{aligned} \text{Also Minimize } z &= -\text{Max}(z^1) \\ &= -(-768/5) \\ &= 768/5 \end{aligned}$$

$$\therefore \text{Minimize } z = 768/5$$

6) ii) Use two-phase method to solve the following LPP.

$$\text{Minimize } z = x_1 + x_2$$

s.t.

$$2x_1 + x_2 \geq 4$$

$$2x_1 + 7x_2 \geq 7$$

$$x_1, x_2 \geq 0$$

Soln:

Step 1: Given LPP is rewritten as,

$$\begin{aligned}\text{Minimize } z &= \text{Max}(-z) \\ &= \text{Max}(z^*)\end{aligned}$$

$$\text{Max } z^* = -x_1 - x_2$$

S.T.

$$2x_1 + x_2 \geq 4$$

$$2x_1 + 7x_2 \geq 7$$

$$\text{where } x_1, x_2 \geq 0$$

Step 2: By introducing non-negative surplus and artificial variable into the standard form of the LPP, becomes.

$$\text{Max } z^* = -x_1 - x_2$$

S.T.

$$2x_1 + x_2 - s_1 + 0s_2 + R_1 = 4$$

$$2x_1 + 7x_2 + 0s_1 - s_2 + R_2 = 7$$

$$\text{where } x_1, x_2, s_1, s_2, R_1, R_2 \geq 0$$

$$n = 6$$

$$m = 2$$

$$n - m = 6 - 2$$

$$= 4$$

$$x_1 = x_2 = s_1 = s_2 = 0$$

$$R_1 = 4$$

$$R_2 = 7$$

PHASE - I

The objective function of the auxiliary LP becomes

$$\text{Max } z^* = -R_1 - R_2$$

C_B	X_B	C_j	0	0	0	0	-1	-1	θ
	X_B		x_1	x_2	s_1	s_2	R_1	R_2	
-1	R_1	4	2	1	-1	0	1	0	4
-1	R_2	7	2	7	0	-1	0	1	1
	Z_j	-11	-4	-8	1	1	-1	-1	
	$C_j - Z_j$	-11	+4	8	-1	-1	0	0	

↑

Here $C_j - Z_j \geq 0 \forall j$

The current BFS is not optimal

So x_2 enters the basis

R_2 leaves the basis

$\therefore$ Pivotal element is 7

First Iteration

C_B	Y_B	C_j	0	0	0	0	-1	0
		x_B	x_1	x_2	s_1	s_2	R_1	
-1	R_1	3	$12/7$	0	$-1 \rightarrow 1/7$	1	1	$3/12 = 1/4$ $7 = 1.75$
0	x_2	1	$2/7$	1	0	$-1/7$	0	$1/2 = 0.5$ $7 = 1.5$
Z_j		-3	$-12/7$	0	1	$-1/7$	-1	
$C_j - Z_j$		-3	$12/7$	0	-1	$1/7$	0	

Here $C_j - Z_j \geq 0 \forall j$

$\therefore$ The current BFS is not optimal

So x_1 enters the basis

R_1 leaves the basis

$\therefore$ Pivotal element is $12/7$

Second Iteration:

C_B	Y_B	C_j	0	0	0	0
		X_B	x_1	x_2	s_1	s_2
0	x_1	$7/4$	1	0	$-7/12$	$1/12$
0	x_2	$1/2$	0	1	$1/6$	$-1/6$
Z_j		0	0	0	0	0
$C_j - Z_j$		0	0	0	0	0

Here $C_j - Z_j \leq 0 \forall j$

The current TBFS is optimal and feasible.

So further, the set of basic variables in the optimal phase-I does not contain artificial variable so the given problem is feasible.

Phase-II:

Here we consider the actual cost associated with the original variables, then the new objective function becomes.

$$\text{Max } z = x_1 + x_2 + 0s_1 + 0s_2$$

The ZBFs for this phase is one obtained at the end of the phase - π .

The iterative simplex table for this phase one.

$$\text{Max } z = -x_1 - x_2 + 0s_1 + 0s_2$$

C_B	Y_B	C_j	-1	-1	0	0
		X_B	x_1	x_2	s_1	s_2
-1	x_2	7/4	1	0	-7/12	1/12
-1	x_2	1/2	0	1	1/6	-1/6
	Z_j	-9/4	-1	-1	5/12	1/12
	$C_j - Z_j$	-9/4	0	0	-5/12	-1/12

Here $C_j - Z_j \leq 0 \forall j$. The current is optimal & feasible.

$$\therefore x_1 = 7/4 ; x_2 = 1/2$$

$$\text{Maximize } z^* = -9/4$$

$$\begin{aligned} \text{Minimize } z &= -\max(z^*) \\ &= -(-9/4) \\ &= 9/4 \end{aligned}$$

6)iii) Use Two-Phase Method to solve the following LP.

$$\text{Maximize } Z = 20x_1 + 10x_2 + 15x_3$$

s.t.

$$8x_1 + 6x_2 + 2x_3 \leq 60$$

$$5x_1 + x_2 + 6x_3 \geq 40$$

$$2x_1 + 6x_2 + 3x_3 \leq 30$$

$$x_1, x_2, x_3 \geq 0$$

Soln:

Step 1:

$$\text{Maximize } Z = 20x_1 + 10x_2 + 15x_3$$

s.t.

$$8x_1 + 6x_2 + 2x_3 \leq 60$$

$$5x_1 + x_2 + 6x_3 \geq 40$$

$$2x_1 + 6x_2 + 3x_3 \leq 30$$

$$\text{where } x_1, x_2, x_3 \geq 0$$

Step 2:

$$\text{Max } Z = 20x_1 + 10x_2 + 15x_3$$

s.t.

$$8x_1 + 6x_2 + 2x_3 + s_1 + 0s_2 + 0s_3 = 60$$

$$5x_1 + x_2 + 6x_3 + 0s_1 - s_2 + 0s_3 + A_1 = 40$$

$$2x_1 + 6x_2 + 3x_3 + 0s_1 + 0s_2 + s_3 + 0R_1 = 30$$

$$\text{where } x_1, x_2, x_3, s_1, s_2, s_3, R_1 \geq 0$$

$$n = 7$$

$$m = 3$$

$$n - m = 4 \text{ variables zero}$$

$$x_1 = x_2 = x_3 = 2 = 0$$

$$s_1 = 60; R_1 = 40; s_3 = 30$$

Initial Iteration:

C_0	Y_0	q	0	0	0	0	0	0	-1	θ
		x_0	x_1	x_2	x_3	s_1	s_2	s_3	R_1	
0	s_1	60	8	6	2	1	0	0	0	$\frac{60}{2} = 30$
-1	R_1	40	5	1	6	0	-1	0	1	$\frac{40}{6} = 6.5$
0	s_3	30	2	6	3	0	0	1	0	$\frac{30}{3} = 10$
z_j		-40	-5	-1	-6	0	1	0	-1	
$C_j - z_j$		-40	5	1	6	0	-1	0	0	

Here $C_j - z_j \geq 0 \forall j$. The current is not optimal

x_3 enters

R_1 leaves

Pivotal element is 6

First Iteration,

C_B	x_B	C_j	0	0	0	0	0	0
		x_B	x_1	x_2	x_3	s_1	s_2	s_3
0	s_1	$280/6$	$18/6$	$34/6$	0	1	$2/6$	0
0	x_3	$40/6$	$5/6$	$1/6$	1	0	$-1/6$	0
0	s_3	$60/6$	$-3/6$	$33/6$	0	0	$3/6$	1
z_j		0	0	0	0	0	0	0
$C_j - z_j$		0	0	0	0	0	0	0

Here $C_j - z_j \leq 0 \forall j$ The current z_{max}
 is optimal and feasible

Phase-II

The iterative simplex table for this phase

$$\text{Max } z = 20x_1 + 10x_2 + 15x_3 + 0s_1 + 0s_2 + 0s_3$$

C_B	X_B	C_j	20	10	15	0	0	0	0
		X_B	x_1	x_2	x_3	s_1	s_2	s_3	
0	s_1	$250/6$	$25/6$	$34/6$	0	1	$2/6$	0	$250/10 = 25$
0	x_3	$40/6$	$5/6$	$1/6$	1	0	$-7/6$	0	$40/15 = 8$
0	s_3	$60/6$	$-3/6$	$33/6$	0	0	$2/6$	1	
	Z_j	100	$25/6$	$15/6$	15	0	$-15/6$	0	
	$C_j - Z_j$	100	$45/6$	$45/6$	0	0	$15/6$	0	

Here $C_j - Z_j \leq 0 \forall j$, The current is optimal

Since $C_j - Z_j \leq 0 \forall j$. The current BFS is not optimal

So, x_1 is entering the basis

s_1 exits the basis

The pivot element is $38/6$

C_B	Y_B	C_j	20	10	15	0	0	0	θ
		Y_D	x_1	x_2	x_3	s_1	s_2	s_3	
20	x_1	$\frac{280}{38}$	1	$34/38$	0	$6/38$	$2/38$	0	40
15	x_3	$20/38$	0	$-11/38$	1	$-5/38$	$-8/38$	0	-
0	s_3	$\frac{526}{38}$	0	$226/38$	0	$3/38$	$20/38$	1	-
Z_j		$5900/38$	20	$356/38$	15	$14/38$	$-80/38$	0	
$C_j - Z_j$		$5900/38$	0	$30/38$	0	$-45/38$	$80/38$	0	

Third Iteration.

CB	YB	C_j	20	10	15	0	0	0
		x_0	x_1	x_2	x_3	s_1	s_2	s_3
20	x_1	6	1	$3/10$	0	$3/20$	0	$-1/10$
15	x_3	6	0	$9/5$	1	$-1/10$	0	$2/5$
0	s_2	16	0	$26/10$	0	$3/20$	1	$38/10$
Z_j		210	20	33	15	$3/2$	0	4
$C_j - Z_j$		210	0	-23	6	$-3/2$	0	-4

Here $C_j - Z_j \leq 0 \forall j$. So current BFS is optimal

$$\therefore x_1 = 6, x_2 = 0, x_3 = 6$$

$$\text{Maximum } Z = 210$$